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COMP30023 – Computer Systems

Application Layer – DNS – Mail

Dr Lachlan Andrew

2025

Recap

- Sockets are the “door” between application layer (user space) and transport layer (kernel space)
- One side opens passively, the other opens actively

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Network addresses

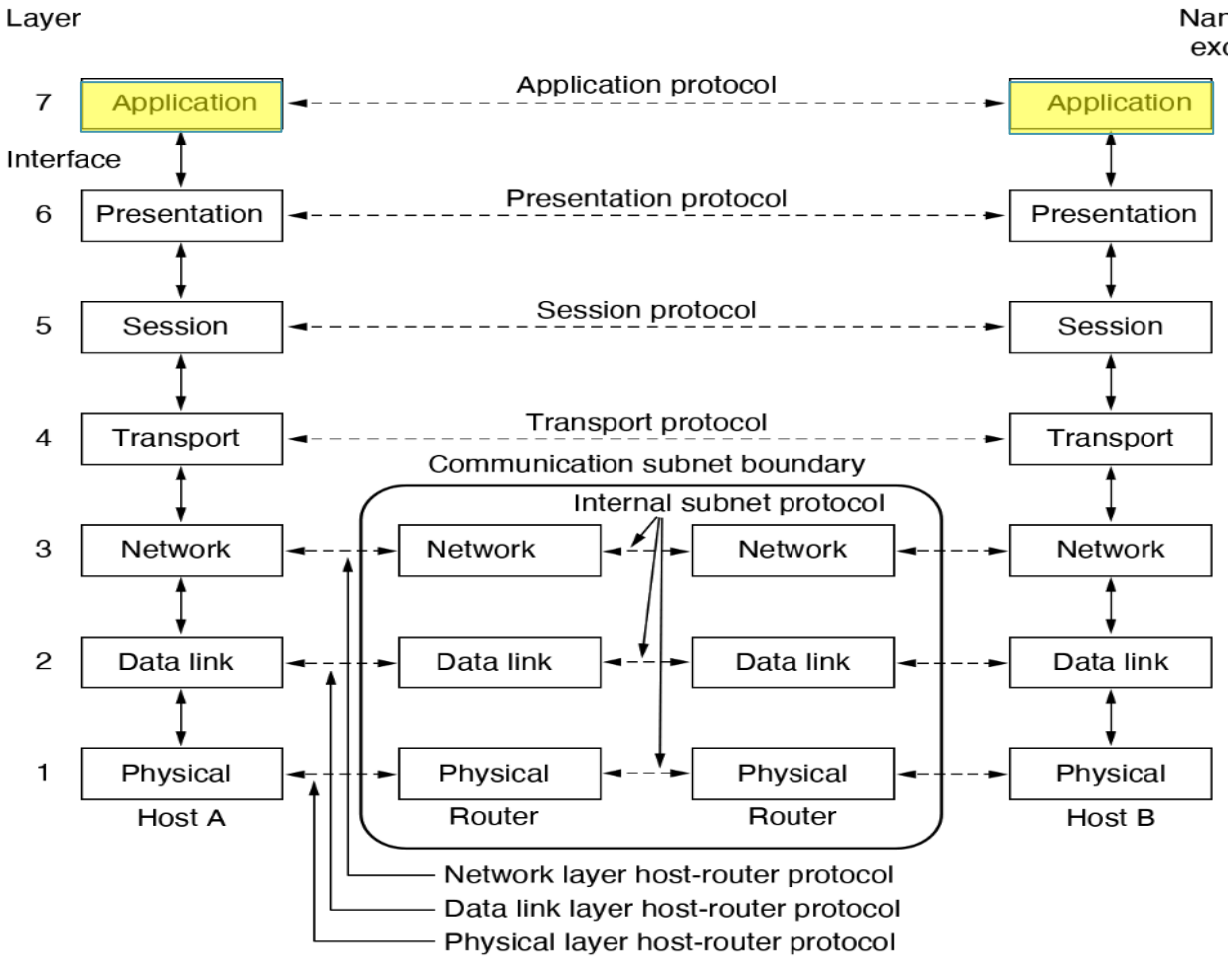
```
    // #define required for VSCode
#define _POSIX_C_SOURCE 200112L
#include <netdb.h>
#include <string.h>
#include <stdio.h>
#include <unistd.h>

int connfd = 0, re = 1, s, n;
struct addrinfo hints, *res, *rp;

memset(&hints, 0, sizeof hints);
hints.ai_family = AF_INET6;
hints.ai_socktype = SOCK_STREAM;
```

```
s = getaddrinfo("localhost",
                "5000", &hints, &res);
for (rp = res; rp != NULL;
            rp = rp->ai_next) {
    connfd = socket(rp->ai_family,
                    rp->ai_socktype,
                    rp->ai_protocol);
    if (connfd == -1) continue;
    if (connect(connfd,
                rp->ai_addr,
                rp->ai_addrlen) != -1
        break;
    close(connfd);
}
write (connfd, "Hello, network!\r\n");
close (connfd);
```

In the stack: Application layer



- Use data
- Tidy up e2e
- Get data e2e
- Tidy up p2p
- Get data p2p

Summary: Application layer

- Domain name service (DNS)
- Email
  - Simple Mail Transfer Protocol (SMTP)
  - Internet Message Access Protocol (IMAP)

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DNS: Mapping names to numbers

- We use “domain names”
  - `facebook.com`
  - `unimelb.edu.au`
  - `student.unimelb.edu.au`
- Sockets want IP addresses
  - 128.250.81.2
  - fe80::69a4:1496:4354:1862
- The Domain Name Service (DNS) maps human-readable names to computer-friendly IP addresses
- Works at the application layer
  - It has to create a socket to find out the IP address
  - Needs to work with a hard-coded IP address to open that socket

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Domain Name System - DNS

- DNS is the technology behind mapping `host.domain.com` to an IP address.
- Four elements comprise the DNS:
  - **Domain name space:** DNS uses a tree-structured name space to identify resources on the Internet.
  - **DNS database:** Each node/leaf in the name space tree names a set of information that is contained in a resource record (RR). The collection of all RRs is organized into a *distributed* database.
  - **Name servers:** Server programs that hold information about a portion of the domain name tree structure and the associated RRs.
  - **Resolvers:** These are programs that extract information from name servers in response to client requests.

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Domain name characteristics

- Domain names:
  - are *not* case sensitive
  - "constituents" separated by "." (*unrelated* to IP address ".")
  - can have up to 63 characters per constituent
  - can have up to 255 chars per path
  - can be internationalised (since 1999) – caused security problems

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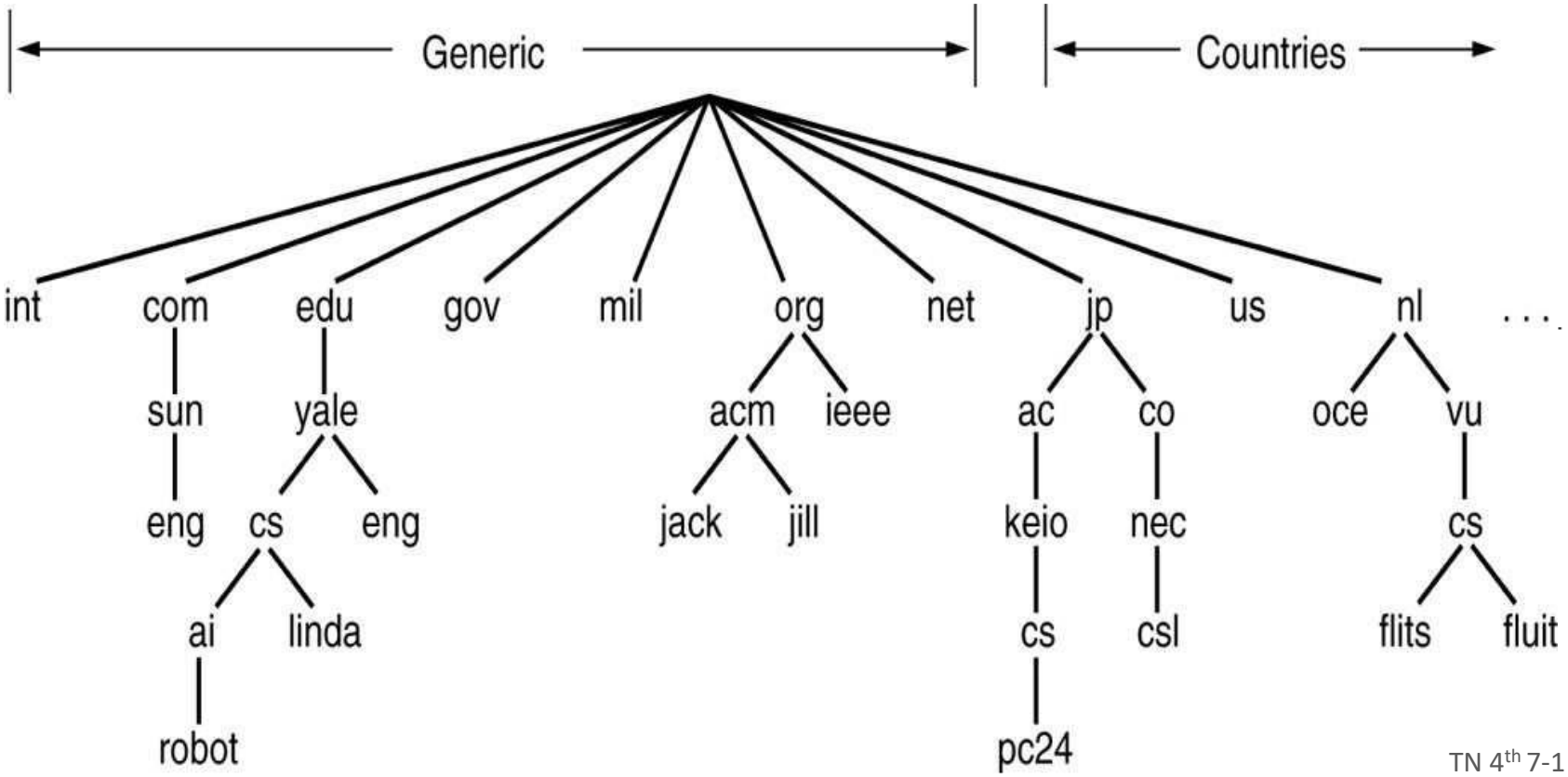
Conceptual division of DNS namespace



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Top-level domains

- The same often followed within country TLDs
- Don't memorise the TLDs
  - Just know they are "Things like .com, .edu"

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No

+ many less common

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Resource Records

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

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Resource Record Example

Absolute domain names ends in a '.'

Relative domain names end in a constituent e.g., .com

```
; Authoritative data for cs.vu.nl
cs.vu.nl.        86400   IN   SOA     star boss (9527,7200,7200,241920,86400)
cs.vu.nl.        86400   IN   MX      1 zephyr
cs.vu.nl.        86400   IN   MX      2 top
cs.vu.nl.        86400   IN   NS      star

star             86400   IN   AAAA    1062::A22:AAE9:51C7:8451
zephyr           86400   IN   A       130.37.20.10
top              86400   IN   A       130.37.20.11
www              86400   IN   CNAME   star.cs.vu.nl
ftp              86400   IN   CNAME   zephyr.cs.vu.nl

flits            86400   IN   A       130.37.16.112
flits            86400   IN   A       192.31.231.165
flits            86400   IN   MX      1 flits
flits            86400   IN   MX      2 zephyr
flits            86400   IN   MX      3 top

rowboat                  IN   A       130.37.56.201
                         IN   MX      1 rowboat
                         IN   MX      2 zephyr

little-sister            IN   A       130.37.62.23

laserjet                 IN   A       192.31.231.216
```

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Multiple addresses for one name

```
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#include <unistd.h>

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struct addrinfo hints, *res, *rp;

memset(&hints, 0, sizeof hints);
hints.ai_family = AF_INET6;
hints.ai_socktype = SOCK_STREAM;
```

```
s = getaddrinfo("localhost",
                "5000", &hints, &res);
for (rp = res; rp != NULL;
                rp = rp->ai_next) {
    connfd = socket(rp->ai_family,
                    rp->ai_socktype,
                    rp->ai_protocol);
    if (connfd == -1) continue;
    if (connect(connfd,
                rp->ai_addr,
                rp->ai_addrlen) != -1)
        break;
    close(connfd);
}
write (connfd, "Hello, network!\r\n");
close (connfd);
```

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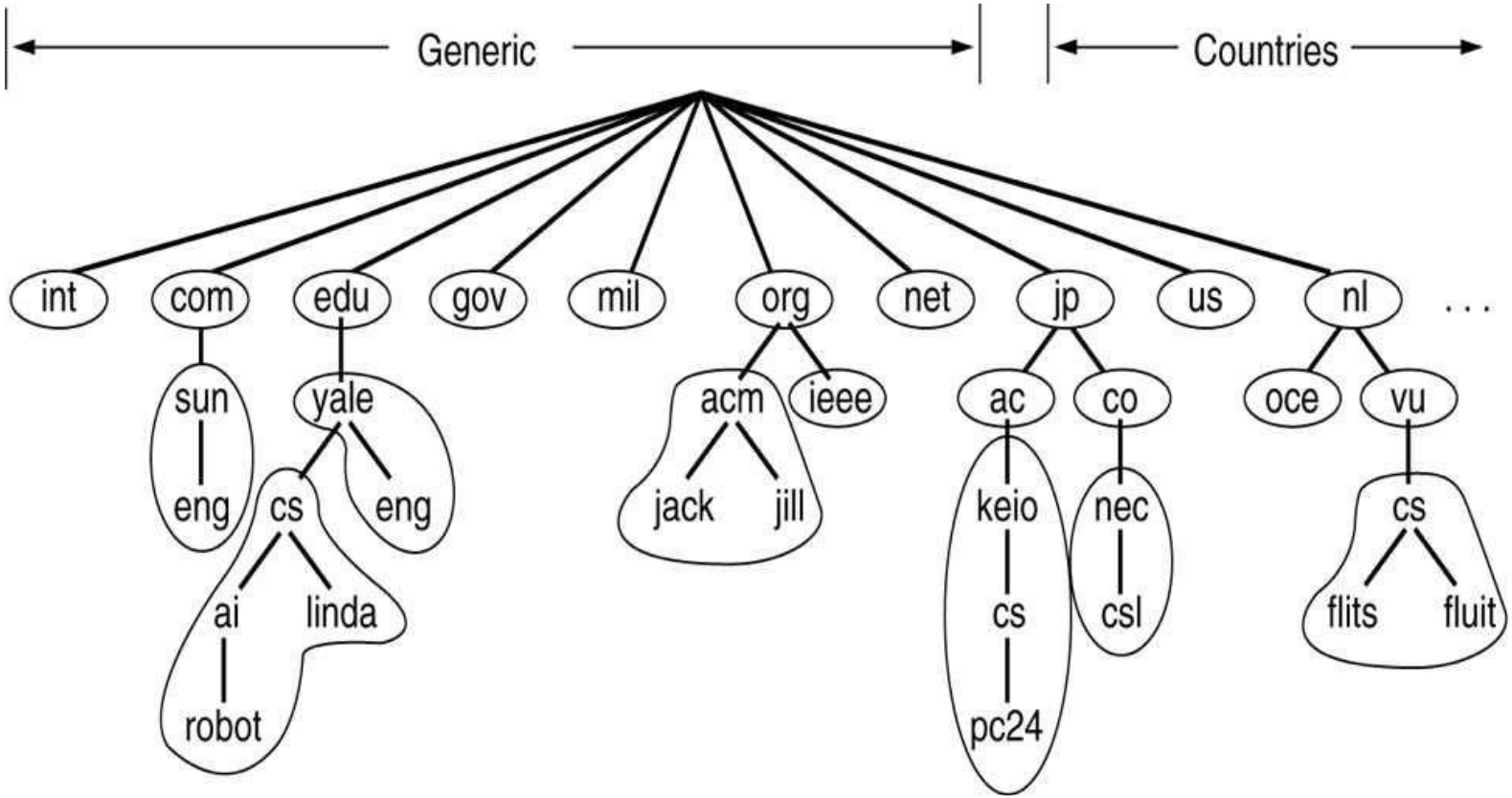
Name server zones



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Name server zones

- Zones:
  - DNS namespace is divided into non-overlapping zones. The name servers are authoritative for that zone.
    - usually two name servers for a zone
  - Name servers are arranged in a hierarchical manner extending from a set of root servers
- Root name servers:
  - The root servers form the authoritative cluster for enquiries. The root servers are contacted by a local name server that cannot resolve name.

Types of name servers

- **Top-level domain DNS servers**: responsible for com, org, net, edu, etc, and all top-level country domains uk, fr, au, jp.
  - Examples include: Network Solutions maintains servers for com; and Educause for edu
- **Other authoritative DNS servers**: organizations DNS servers, providing authoritative hostname to IP mappings for organizations servers (e.g., Web, mail).
  - Can be maintained by the organization itself or service provider.
- **Local DNS server**: Typically, each ISP (residential ISP, company, university) has a "default name server" which handles DNS queries
  - Returns cached value if one exists
  - Otherwise, acts as proxy, and forwards the request up the query hierarchy

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Resolving a query

- A resolver client asks the local DNS for the domain to IP mapping:
  - if answer is known by the local DNS, then it sends the answer.
  - if answer is not known, then the local DNS queries up the hierarchy to the top level (root) DNS for the domain and then relays the answer to the resolver client.
- Queries are subject to timers to avoid longer than necessary response times.

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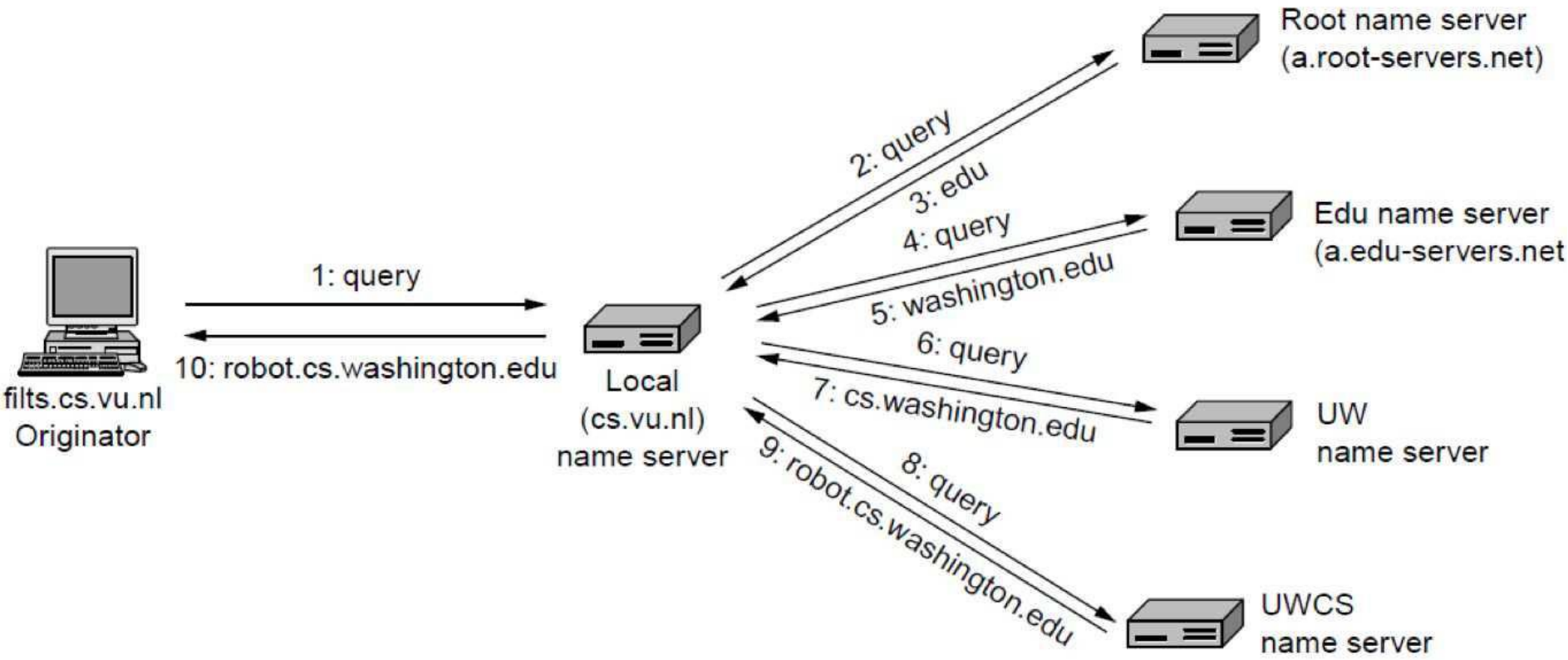
Example Resolver Query



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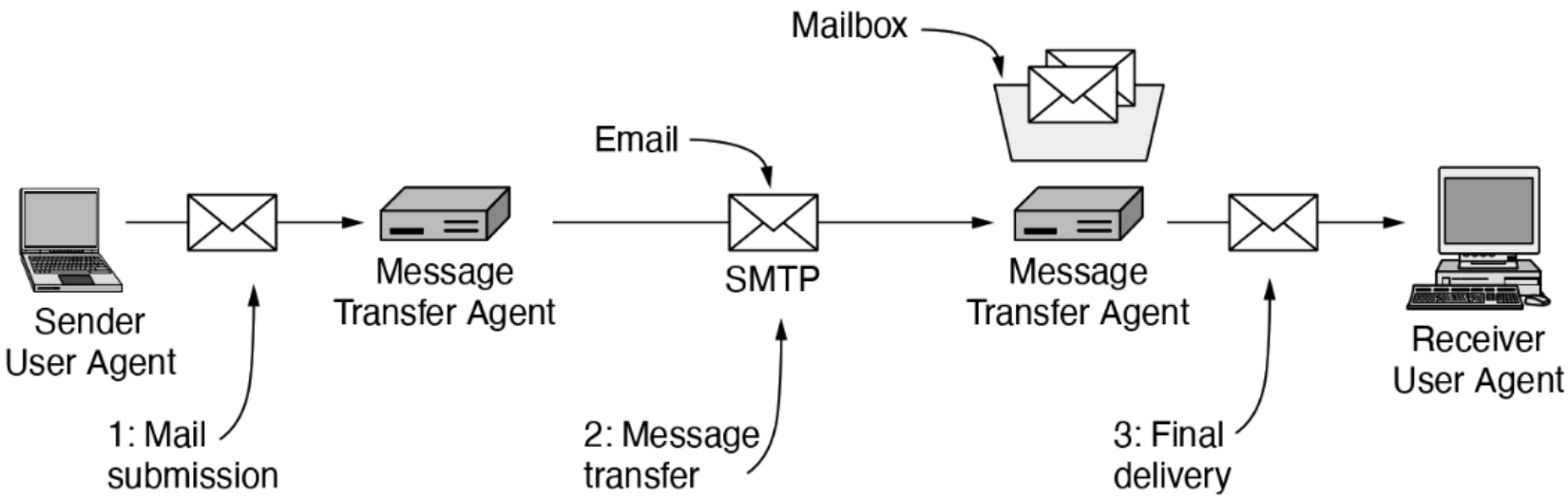
Email services and architecture

- Started in the 1960s
  - Has evolved a lot!
- Standards for Internet email are mainly based on two RFCs
  - RFC 2821 (transmission)
  - RFC 2822 (message format)

  (Don't remember these numbers, but take a look at the RFCs)
- Architecture and Services
  - User agents (UA's/ MUA's)
    - allow user to read and send email
  - Message transfer agents (MTA's)
    - transport messages from source - destination

Email services and architecture



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There may be SMTP relays between the sender's MTA and the receiver's
https://blog.mailchannels.com/what-is-an-smtp-relay-service

User agent (mail program)

- Basic functions:
  - compose, report, display, dispose
- Envelope and contents:
  - encapsulation of transport related information
- Header
  - user agent control info
- Body
  - for human recipient
- User must provide message, destination, optional other parameters
- Addressing scheme user@dns-address

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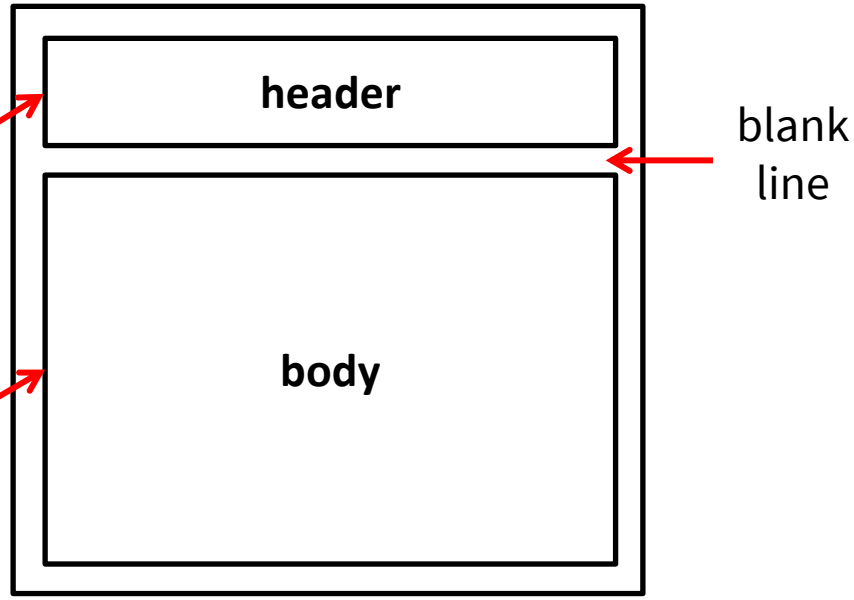
Mail Message Format

SMTP: protocol for exchanging email messages
RFC 2822: standard for text message format:

- **header lines, e.g.**
  To:
  From:
  Subject:
- **body**
  the "message", ASCII characters only



Message Header Fields

Headers the user usually sees

To:	Cc:	Bcc:
From:	Date:	Subject:

Headers the user usually doesn't see.

Message-Id:	In-Reply-To:	References:
Reply-To:	Sender:	Return-Path:
Keywords:	Received:	

What are they for?

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SMTP – Simple Message Transfer Protocol

- SMTP uses TCP to reliably transfer email message from client to server, default port 25
- Typically direct transfer: sending server to receiving server
- Three phases of transfer (1) handshaking (greeting), (2) transfer of messages, and (3) closure
- Command/response interaction: commands in ASCII text and response consists of status code and phrase
- Messages must be in 7-bit ASCII

Transfer protocol – analogous to HTTP

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SMTP – Interaction Example

Note: many back-and-forth exchanges.

Slow on modern networks, where latency is larger than serialization delay.

Modern approach of one header has lower delay.

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

MIME – Multipurpose Internet Mail Extensions

- In the early days of email, messages were in English and used only ASCII - RFC 822 reflects these simple constraints. In time, the limitations of RFC822 became clear:
  - other language requirements
  - alternative message content type (audio/images)
- MIME has 5 additional message headers:
  - MIME-Version: identifies the MIME version
  - Content-Description: human readable describing contents
  - Content-Id: unique identifier
  - Content-Transfer-Encoding: how body is wrapped for transmission
  - Content-Type: type and format of content

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MIME – Content-Types

Type	Example subtypes	Description
text	plain, html, xml, css	Text in various formats
image	gif, jpeg, tiff	Pictures
audio	basic, mpeg, mp4	Sounds
video	mpeg, mp4, quicktime	Movies
model	vrml	3D model
application	octet-stream, pdf, javascript, zip	Data produced by applications
message	http, rfc822	Encapsulated message
multipart	mixed, alternative, parallel, digest	Combination of multiple types

e.g., text/plain, multipart/alternate

Message Transfer & Access

- Transfer
  - SMTP: delivery/storage to receiver's server
- Delivery
  - Local
  - POP3 : Post Office Protocol; authorization (agent – server) and download
  - IMAP: Internet Mail Access Protocol; more features (more complex); provides for the manipulation of stored messages on server
  - HTTP: gmail, Hotmail, Yahoo! Mail, etc.

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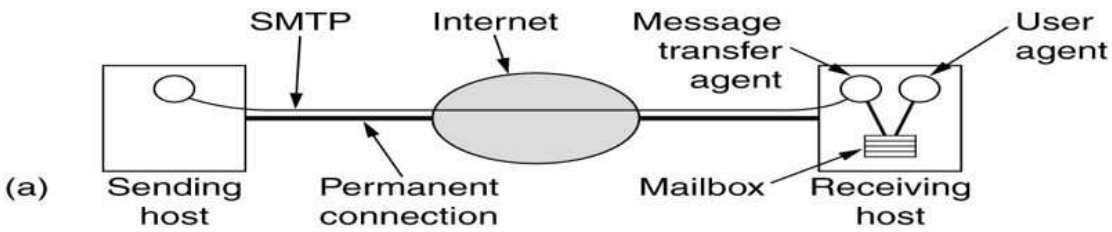
Receiving Mail: Local vs Remote



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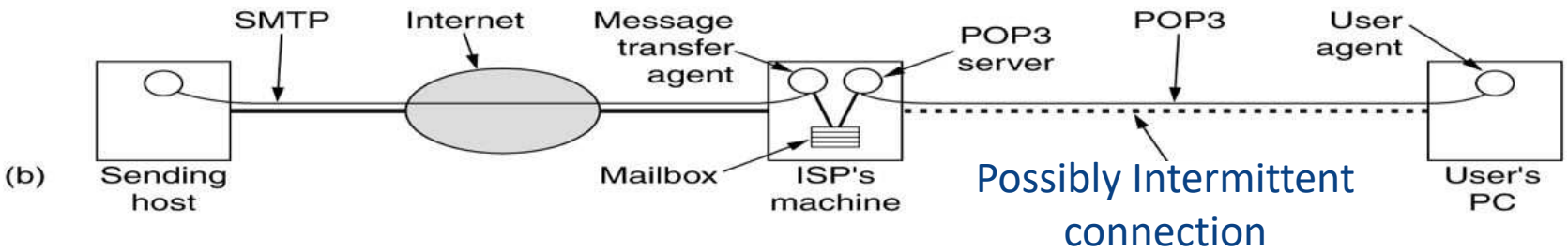

a) Sending and reading mail when the receiver has a permanent Internet connection and the user agent runs on the same machine as the message transfer agent (now rare).

b) Current case: notebook/PC/phone is not an MTA.

IMAP – Internet Message Access Protocol

- IMAP allows users to query a MTA
- IMAP keeps user state across sessions.
  - Retain mailbox contents online (server) and allow manipulation of online and offline messages and mailbox folders
  - Implications of server infrastructure to support high volume of IMAP users. This implies storage projections by the provider, and hence limitations.

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Example from RFC 3501

C: <open connection>

S: * OK IMAP4rev1 Service Ready

C: a001 login mrc secret

S: a001 OK LOGIN completed

Authorization phase

C: a002 select inbox

S: * 18 EXISTS

S: * FLAGS (\Answered \Flagged \Deleted \Seen \Draft)

S: * 2 RECENT

S: * OK [UNSEEN 17] Message 17 is the first unseen message

S: * OK [UIDVALIDITY 3857529045] UIDs valid

S: a002 OK [READ-WRITE] SELECT completed

Example from RFC 3501 cont'd

```
C:  a003 fetch 12 full
S:  * 12 FETCH (FLAGS (\Seen) INTERNALDATE "17-Jul-1996 02:44:25 -0700"
    RFC822.SIZE 4286 ENVELOPE ("Wed, 17 Jul 1996 02:23:25 -0700 (PDT)"
    "IMAP4rev1 WG mtg summary and minutes"
    (("Terry Gray" NIL "gray" "cac.washington.edu"))
    (("Terry Gray" NIL "gray" "cac.washington.edu"))
    (("Terry Gray" NIL "gray" "cac.washington.edu"))
    ((NIL NIL "imap" "cac.washington.edu"))
    ((NIL NIL "minutes" "CNRI.Reston.VA.US")
    ("John Klensin" NIL "KLENSIN" "MIT.EDU")) NIL NIL
    "<B27397-0100000@cac.washington.edu>")
    BODY ("TEXT" "PLAIN" ("CHARSET" "US-ASCII") NIL NIL "7BIT" 3028
    92))
S:  a003 OK FETCH completed
```

Example from RFC 3501 cont'd

```
C:  a004 fetch 12 body[header]
S:  * 12 FETCH (BODY[HEADER] {342}
S:  Date: Wed, 17 Jul 1996 02:23:25 -0700 (PDT)
S:  From: Terry Gray <gray@cac.washington.edu>
S:  Subject: IMAP4rev1 WG mtg summary and minutes
S:  To: imap@cac.washington.edu
S:  Cc: minutes@CNRI.Reston.VA.US, John Klensin <KLENSIN@MIT.EDU>
S:  Message-Id: <B27397-0100000@cac.washington.edu>
S:  MIME-Version: 1.0
S:  Content-Type: TEXT/PLAIN; CHARSET=US-ASCII
S:
S:  )
S:  a004 OK FETCH completed
C   a005 store 12 +flags \deleted
S:  * 12 FETCH (FLAGS (\Seen \Deleted))
S:  a005 OK +FLAGS completed
C:  a006 logout
S:  * BYE IMAP4rev1 server terminating connection
S:  a006 OK LOGOUT completed
```

And finally...

The Register® OFF-PREM ▾ ON-PREM ▾ SOFTWARE ▾ SECURITY OFFBEAT ▾ VENDOR VOICE ▾

{* BOOTNOTES *}

Microsoft staff giggle beneath the weight of a 52,000-person Reply-All email storm

Team Redmond stokes the flames as an exercise in black humor

Jeffrey Snover
@jsnover

All you need to do is:
1) REPLY-ALL saying "Don't REPLY-ALL you idiots"
2) REPLY-ALL saying "Please take me off this list"
3) Repeat

8:58 AM · Mar 27, 2020

Acknowledgement

- The slides were based on slides prepared by Chris Culnane, based on material developed previously by: Michael Kirley, Zoltan Somogyi, Rao Kotagiri, James Bailey and Chris Leckie.
- Some of the images included in the notes were supplied as part of the teaching resources accompanying the text books listed on the previous slides.
  - (And also) Computer Networks, 6th Edition, Tanenbaum A., Wetherall. D. https://ebookcentral.proquest.com/lib/unimelb/detail.action?docID=6481879
- Textbook Reference: Sections 2.3, 2.4, 2.6, and related topics on pp.199-210

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- The following is non-examinable material
- It has been removed from course.
- It is here in case you find it a useful reference

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Inserting records into DNS

- Example: new start-up Network Utopia
- Register name networkuptopia.com at DNS registrar (e.g., Network Solutions)
  - provide names, IP addresses of authoritative name server (primary and secondary)
  - registrar inserts two RRs into com TLD server:
    - (networkutopia.com, dns1.networkutopia.com, NS)
    - (dns1.networkutopia.com, 212.212.212.1, A)
  - create authoritative server:
    - Type A record for www.networkuptopia.com;
    - Type MX record for networkutopia.com

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DNS Security

- No security in original design
  - DNS spoofing
  - DNS flooding
- Solutions
  - DNSSEC
  - Root signing

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POP3 – Post Office Protocol

- Three states of a POP3 transaction
  - Authorisation
  - Transactions
  - Update
- Syntax
  - USER / PASS
  - LIST
  - RETR / DELE
  - QUIT (update)
- Issue: “download and delete” mode does not allow messages to be re-read.

POP3 – Post Office Protocol

authorization phase

- client commands:
  - `user:` declare username
  - `pass:` password
- server responses
  - `+OK`
  - `-ERR`

transaction phase, client:

- `list:` list message numbers
- `retr:` retrieve message by number
- `dele:` delete
- `quit`

```
S: +OK POP3 server ready
C: user alice
S: +OK
C: pass hungry
S: +OK user successfully logged on
```

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: <message 1 contents>
S: .
C: dele 1
C: retr 2
S: <message 1 contents>
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

Streaming

- 40% of internet download traffic is streaming (video, audio)
  - Not necessarily 40% of core traffic, due to caching
- WebSockets (ws://… and wss://…)
  - Multiplexing full-duplex channels over one TCP connection
  - Everything on TCP port 80 or 443 (HTTP/HTTPS) to pass firewalls
  - Unlike HTTP, doesn't require client to keep requesting. → streaming
- RTP + RTCP/RTSP (Real Time [Control/Streaming] Protocol)
  - RTP uses a playback buffer to deliver packets with the same spacing as they were sent, undoing network jitter
  - RTCP Monitors delays, adapts video coding rate to available capacity
  - RTSP provides play/record/pause services
- Real Time Messaging Protocol (RTMP, used by Flash)

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Recap

Email

- SMTP – main email protocol, push-based
- POP/IMAP – pull-based for collection by the final recipient
- Very “chatty” protocols, with many round-trip exchanges